BASE CHANGE IS LEFT ADJOINT TO RESTRICTION OF SCALARS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof ends with the author's sentence that the verification of the two constructions being mutually inverse is omitted; that omission is routine, the constructions themselves are fully written, and the ending sentence is kept verbatim inside the span.

Editorial note. The counted spans contain no \ref and no \cite, so the card carries no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- The S-module structure on the base change of an R-module. Used at the construction of beta in the counted proof (lines 2698-2699). Makes the domain of the adjunction map an S-module, so that an S-module map out of it can be tested on the tensor product.
- The pure tensor m tensor 1 as the natural image of an element. Used at proof line 2700. Gives the reverse construction beta'(m) = beta(m tensor 1).

Lemma 0.1. *Let $R \to S$ be a ring map. The functors $Mod_S \to Mod_R$, $N \mapsto N_R$ (restriction) and $Mod_R \to Mod_S$, $M \mapsto M \otimes_R S$ (base change) are adjoint functors. In a formula*

$$\mathrm{Hom}_R(M, N_R) = \mathrm{Hom}_S(M \otimes_R S, N)$$

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Proof. If $\alpha : M \to N_R$ is an R-module map, then we define $\alpha' : M \otimes_R S \to N$ by the rule $\alpha'(m \otimes s) = s\alpha(m)$. If $\beta : M \otimes_R S \to N$ is an S-module map, we define $\beta' : M \to N_R$ by the rule $\beta'(m) = \beta(m \otimes 1)$. We omit the verification that these constructions are mutually inverse. $\square$